Image Processing and Pattern Recognition Syllabus (3-1-2)

Theory + Practical **Total Course Hours: 45 Hrs**

Course Contents:

1. Introduction to Digital Processing **[4 Hour]**

- 1.1. The origin of Digital Image Processing
- 1.2. Examples of Fields that use Digital Processing
- 1.3. Fundamental Steps in Image Processing Systems
- 1.4. Elements of Digital Image Processing Systems
- 1.5. Image Sampling and Quantization
- 1.6. Some Basic relationship like Neighbor's, Connectivity, Distance Measures between Pixels
- 1.7. Elements of Visual Perception

2. Image Enhancement in Spatial Domain **[7 Hour]**

- 2.1. Some basic Gray level Transformations
  - 2.1.1. Point operations
  - 2.1.2. Contrast Stretching
  - 2.1.3. Thresholding
  - 2.1.4. Digital Negative
  - 2.1.5. Intensity level slicing
  - 2.1.6. Bit plane Slicing
- 2.2. Histogram Processing and Equalization
- 2.3. Enhancement Using Arithmetic and Logic Operations
- 2.4. Basics of Spatial Filters
- 2.5. Smoothening and Sharpening Spatial Filters
  - 2.5.1. Averaging
  - 2.5.2. Media Filtering
  - 2.5.3. Spatial Low Pass
  - 2.5.4. High Pass Filtering
  - 2.5.5. Magnification by replication and Interpolation
  - 2.5.6. Combining Spatial Enhancement Methods

3. Image Enhancement in the Frequency Domain **[6 Hour]**

- 3.1. Introduction to Fourier Transform and the frequency Domain
- 3.2. Computing and Visualizing the 2D DFT
- 3.3. Smoothening Frequency Domain Filters
- 3.4. Sharpening Frequency Domain Filters
- 3.5. Other Image Transforms
  - 3.5.1. Hadamard Transform
  - 3.5.2. Haar Transform
  - 3.5.3. Discrete Cosine Transform
- 3.6. Fast Fourier Transform

4. Image Restoration **[4 Hour]**

4.1. A model of the image degradation/Restoration Process
4.2. Noise Models Restoration in the Presence of Noise only Spatial Filtering
4.3. Periodic Noise Reduction by Frequency Domain Filtering

5. Image Compression **[6 Hour]**

5.1. Coding Redundancy
  5.1.1. Huffman Coding
5.2. Interpixel Redundancy
  5.2.1. Run length Coding
5.3. Psychovisual Redundancy
5.4. Image Compression Models
5.5. Lossless and Lossy Compressions
  5.5.1. Predictive Coding

6. Introduction to Morphological Image Processing **[4 Hour]**

6.1. Logic Operation involving binary images
6.2. Dilation and Erosion
6.3. Opening and Closing

7. Image Segmentation **[7 Hour]**

7.1.Detection of Discontinuities
7.2. Edge linking and Boundary Detection
7.3. Thresholding
7.4. Region Based Segmentation

8. Representation and Description **[3 Hour]**

8.1. Introduction to some descriptors
  8.1.1. Chain Codes
  8.1.2. Signatures
  8.1.3. Shape Numbers
  8.1.4. Fourier Descriptors

9. Object Recognition **[3 Hour]**

9.1.Pattern and Pattern Classes
9.2. Decision –Theoretic Methods
9.3. Overview of Neural Network in Image Processing

10. Pattern Recognition **[1 Hour]**

10.1. Overview of Pattern recognition

Laboratory: Students has to be able to write codes regarding image enhancements